Game Theory

Final Examination | Fall 2023

Prof. R. Alvarez

Time limit: 170 minutes • Total: 100 points

Name: _____ Student ID: _____

Instructions

- Closed book and closed notes. Two handwritten $8.5'' \times 11''$ sheets of notes (both sides) are allowed.
- No calculators or electronic devices. Leave answers as exact fractions or simplified surds.
- The exam is cumulative. Show your work; unsupported answers receive little credit.
- In every bimatrix, the first entry is Row's (Player 1's) payoff and the second is Column's (Player 2's).

Problem	1	2	3	4	5	Total
Points	18	22	20	22	18	100
Score						

Problem 1 A 3×3 Zero-Sum Game [18 points]

Entries are payments from Column to Row (Row maximizes).

	c_1	c_2	c_3
r_1	4	-1	5
r_2	-2	3	-1
r_3	0	0	-1

(a) [4] Show that r_3 is not dominated by any pure strategy of Row, but is strictly dominated by $\alpha r_1 + (1-\alpha) r_2$. Find all α for which this holds.

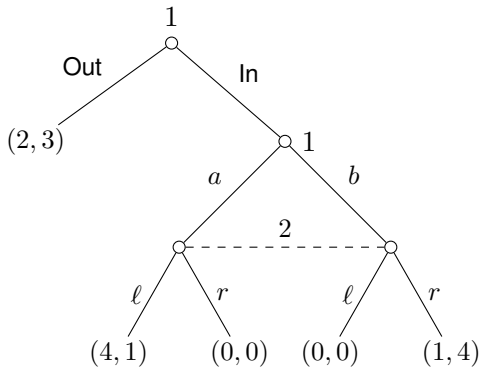
(b) [3] After deleting r_3, show that one of Column's strategies becomes dominated. Explain why that column was *not* dominated in the original game.

(c) [7] Solve the remaining 2×2 game: find the value and both players' optimal strategies.

(d) [4] State optimal strategies for both players in the original 3×3 game, and verify directly (by computing the relevant expected payoffs) that each player's strategy guarantees the value.

Problem 2 An Extensive Game with a Simultaneous-Move Subgame [22 points]

Player 1 first chooses Out or In. After Out the game ends with payoffs $(2, 3)$. After In, Players 1 and 2 play the simultaneous-move game shown on the right (Player 2 does not observe Player 1's choice of a or b).



Subgame after In:

	ℓ	r
a	4, 1	0, 0
b	0, 0	1, 4

(a) **[5]** List each player's pure strategies and write the normal form of the whole game.

(b) **[9]** Find all subgame-perfect equilibria (SPE), including any that use mixed strategies in the subgame. For each, give the outcome and payoffs.

(c) **[5]** Find a pure-strategy Nash equilibrium of the whole game that is *not* subgame perfect. Explain precisely why it fails subgame perfection.

(d) **[3]** Suppose Player 2 could publicly and irrevocably commit to play r before Player 1 moves. Would Player 2 want to? Compare with Player 2's payoffs in the SPE of part (b).

Problem 3 Repeated Prisoner's Dilemma [20 points]

The following stage game is played in periods $t = 1, 2, \ldots$; both players discount future payoffs by $\delta \in (0, 1)$ and observe all past actions.

	C	D
C	6, 6	1, 8
D	8, 1	3, 3

(a) **[3]** Find the unique Nash equilibrium of the stage game, and explain why the game repeated a *finite* number T of times has a unique SPE.

(b) **[6]** Consider the grim-trigger strategy: play C in period 1 and continue to play C as long as (C, C) has been played in every previous period; otherwise play D forever. Find the smallest δ for which both players using grim trigger is an SPE. Check both the cooperative and the punishment phases.

(c) **[7]** Now consider a *k-period punishment* strategy: play C on the cooperative path; after any deviation from the prescribed action profile (including a deviation during a punishment phase), both players play D for the next k periods and then return to C. Find the smallest δ for which this profile is an SPE when $k = 1$ and when $k = 2$.

(d) [4] Rank the three thresholds from parts (b) and (c) and explain the ranking in one or two sentences.

Problem 4 A Bayesian Game [22 points]

Player 1 (Row) is uncertain about Player 2's preferences. Player 2 is of type t_1 with probability μ and type t_2 with probability $1-\mu$; Player 2 knows her type, Player 1 does not. Both players move simultaneously. Player 1's payoffs do not depend on the type; the payoff tables are:

Type t_1 (prob. μ)

	L	R
U	5, 2	0, 0
D	1, 0	2, 1

Type t_2 (prob. $1-\mu$)

	L	R
U	5, 0	0, 1
D	1, 0	2, 3

(a) [3] Describe each player's set of pure strategies. Show that type t_2 has a strictly dominant action.

(b) [7] For each $\mu \in [0,1]$, find all pure-strategy Bayesian Nash equilibria (BNE).

(c) [8] Let $\mu = \frac{1}{2}$. Find a BNE in which Player 1 and type t_1 both randomize. Give all mixing probabilities.

(d) [4] For $\mu = \frac{1}{2}$, compute Player 1's expected payoff in every BNE you found. Which equilibrium does Player 1 prefer?

Problem 5 Short Answers [18 points]

Each part is worth 3 points. Give a brief justification or computation.

(a) True or False: In a finite two-player zero-sum game, every optimal (maximin) strategy of Row is a best response to every optimal (minimax) strategy of Column.

(b) True or False: Every Nash equilibrium of a finite extensive-form game with perfect information is subgame perfect.

(c) In the stage game of Problem 3, the players alternate forever between (C,C) in odd periods and (D,D) in even periods, starting with (C,C) in period 1. With $\delta = \frac{1}{2}$, compute Player 1's *average* (normalized) discounted payoff $(1-\delta)\sum_{t\geq 1}\delta^{t-1}u_t$.

(d) Player 1 has payoffs $\begin{pmatrix} 3 & 1 & 0 \\ 0 & 2 & 4 \end{pmatrix}$ (rows T, B; columns L, C, R chosen by Player 2). Compute Player 1's minmax payoff when Player 2 is restricted to pure strategies, and when Player 2 may mix.

(e) In a Bayesian game, Player 1 has 3 types and 2 actions, and Player 2 has 2 types and 3 actions. How many pure-strategy profiles does the game have?

(f) True or False: In an infinitely repeated game, an action that is strictly dominated in the stage game is never played on the equilibrium path of any SPE.

— End of examination —